Decoding Preference Optimization in Financial Domains: A High-Granularity Analysis of Hyperparameter Recalibration and Structural Anomalies across Aligned LLMs

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Abstract

This report delivers an exhaustive empirical and theoretical investigation into the application of Direct Preference Optimization (DPO) for aligning Large Language Models (LLMs) as deterministic, risk-averse quantitative financial assistants. Utilizing a specialized multi-factor statistical matrix token-space, we benchmark four contemporary architectures: Meta-Llama 3 8B Instruct, Google Gemma 2 9B Instruct, Alibaba Qwen 2.5 7B Instruct, and DeepSeek-R1-Distill-Qwen-7B. Our investigation reveals a systemic structural vulnerability to preference overfitting during initial multi-epoch optimization horizons, characterized by a premature saturation of reward accuracy and subsequent collapse of downstream generalization capabilities driven by non-vanishing gradient vectors. We formalize and execute a robust hyperparameter regularization workflow—underpinning learning rate attenuation, epoch contraction, and Kullback-Leibler (KL) penalty amplification—to restore programmatic robustness. Furthermore, we dissect the unique structural and cognitive deviations displayed by each model post-alignment, focusing heavily on a catastrophic token-loop leak and identity boundary collision observed within the reasoning-distilled DeepSeek geometry.

1 Introduction

The implementation of deep autoregressive language models in quantitative finance has shifted from qualitative sentiment extraction to the direct execution of structured risk analysis. Financial time series and statistical asset configurations are characterized by an extremely low signal-to-noise ratio, non-stationarity, and rapid regime alterations. Consequently, deploying Large Language Models (LLMs) to interpret quantitative matrices requires an unprecedented level of behavioral determinism.

Standard foundational models, even those subjected to extensive Instruction Fine-Tuning (IFT), consistently display systemic biases that are highly counterproductive in financial risk management. Chief among these are sycophancy (the tendency to validate a user's erroneous or highly emotional investment propositions) and optimism bias (the systematic underestimation of tail-risk parameters). When presented with retail trading euphoria or social media sentiment (e.g., speculative momentum patterns driven by platforms such as Reddit), unaligned models routinely abandon mathematical constraints to mirror user sentiment.

To resolve this behavioral volatility, this report establishes a rigorous framework leveraging Direct Preference Optimization (DPO) to enforce absolute quantitative discipline. We map statistical outputs—specifically short-term momentum vectors, moving average trend position coordinates, and historical annualized volatility bounds—into a paired preference space. The objective function forces the policy network to anchor its analytical outputs solely on empirical model matrices, actively penalizing behavioral confirmation and qualitative hallucination.

2 Theoretical Core and Gradient Dynamics of DPO

Traditional alignment frameworks historically relied on Reinforcement Learning from Human Feedback (RLHF) via asymmetric multi-stage optimization pipelines. These workflows mandate the training of an independent scalar Reward Model $\mathcal{R}_\phi(x, y)$, followed by policy updates using proximal policy optimization (PPO)—a process notoriously unstable due to high computational overhead and complex hyperparameter sensitivities in actor-critic setups.

Direct Preference Optimization bypasses explicit reward modeling by exploiting an analytical mapping between the optimal policy and the underlying reward function. Under the Bradley-Terry preference model, the probability that a prompt context x yields a preferred response y_w over a dispreferred response y_l is formalized as:

$$P(y_w \succ y_l|x) = \sigma\left(\mathcal{R}^*(x, y_w) - \mathcal{R}^*(x, y_l)\right) \tag{1}$$

Where σ represents the standard logistic function. By leveraging the closed-form solution of the maximum likelihood objective with a Kullback-Leibler (KL) divergence constraint against a reference policy π_{ref}, the latent reward parameter can be mathematically substituted. The generalized DPO objective function optimized during our fine-tuning sequence is expressed as follows:

$$\mathcal{L}_{DPO}(\pi_\theta; \pi_{ref}) = -\mathbb{E}_{(x,y_w,y_l)\sim\mathcal{D}}\left[\log\sigma\left(\beta\log\frac{\pi_\theta(y_w|x)}{\pi_{ref}(y_w|x)} - \beta\log\frac{\pi_\theta(y_l|x)}{\pi_{ref}(y_l|x)}\right)\right] \tag{2}$$

Here, β operates as a critical scaling regularizer that governs the magnitude of the KL-divergence penalty. To truly diagnose why preference models overfit, we must derive the exact gradient of the DPO loss function with respect to the parameters θ. Let the implicit reward estimated by the model be defined as:

$$\hat{r}_\theta(x, y) = \beta\log\frac{\pi_\theta(y|x)}{\pi_{ref}(y|x)} \tag{3}$$

The gradient vector can be analytically structured as:

$$\nabla_\theta\mathcal{L}_{DPO}(\pi_\theta) = -\mathbb{E}_{(x,y_w,y_l)}\left[\beta\cdot\sigma\left(\hat{r}_\theta(x, y_l) - \hat{r}_\theta(x, y_w)\right)\left(\nabla_\theta\log\pi_\theta(y_w|x) - \nabla_\theta\log\pi_\theta(y_l|x)\right)\right] \tag{4}$$

This derivation reveals the foundational mechanics of the optimization framework, pointing to a dual-component dynamic:

1. **The Classifying Multiplier:** The term $(\nabla_\theta \log \pi_\theta(y_w|x) - \nabla_\theta \log \pi_\theta(y_l|x))$ acts as a directional force, increasing the log-likelihood of the preferred tokens while depressing the dispreferred pathways.

2. **The Non-Vanishing Error Weight:** The scalar multiplier $\sigma\left(\hat{r}_\theta(x, y_l) - \hat{r}_\theta(x, y_w)\right)$ functions as an implicit weight proportional to the model's error.

Crucially, because the logistic function $\sigma(z)$ only approaches zero asymptotically as $z \to -\infty$, the gradient weight **never vanishes** even after the model achieves 100% classification accuracy between y_w and y_l. As long as the mathematical margin $(\hat{r}_\theta(x, y_w) - \hat{r}_\theta(x, y_l))$ is finite, the optimizer will continue to ruthlessly alter the model parameters θ to push the probabilities of the preferred tokens to absolute certainty.

3 The Cross-Architectural Baseline Matrix Evaluation

To document the structural contrast between standard preference exploitation and regularized generalization, the experimental baseline captures metrics across both unregularized (3.0 Epochs) and regularized (1.0 Epoch) policy states. Table 1 presents the complete, consolidated parameters spanning all evaluated model families, incorporating the exact empirical step metrics extracted from the hardware logs.

Table 1: Comprehensive DPO Alignment Metrics: Unregularized vs. Calibrated Regimes

Model Architecture	Alignment State	Final Loss	Grad Norm	Reward Accuracy	Reward Margin	Compute Time (s)
Qwen 2.5 7B Instruct	Unregularized (3.0 Epochs)	0.0005	0.018	100.00%	7.8000	688.10
Qwen 2.5 7B Instruct	Calibrated (1.0 Epoch)	0.6404	14.66	84.38%	0.1121	234.70
Meta-Llama 3 8B Instruct	Unregularized (3.0 Epochs)	0.0004	0.016	100.00%	7.9580	712.80
Meta-Llama 3 8B Instruct	Calibrated (1.0 Epoch)	0.5852	14.73	99.38%	0.2315	241.70
Google Gemma 2 9B Instruct	Unregularized (3.0 Epochs)	0.0002	0.011	100.00%	8.1240	1142.50
Google Gemma 2 9B Instruct	Calibrated (1.0 Epoch)	0.4286	16.39	100.00%	0.6343	392.50
DeepSeek-R1-Distill-7B	Unregularized (3.0 Epochs)	0.0012	0.045	100.00%	6.9120	724.30
DeepSeek-R1-Distill-7B	Calibrated (1.0 Epoch)	0.6779	9.466	68.75%	0.0333	237.90

3.1 Analysis of Cross-Regime Data Trajectories

The numerical matrix in Table 1 provides robust empirical validation of the non-vanishing gradient theory. In the unregularized state, all baseline architectures experienced near-zero loss convergence (≤ 0.0012) paired with heavily suppressed gradient norms (≤ 0.045). This indicates that the parameters were locked into local over-optimization sinks.

By contracting the training horizon to exactly 1.0 epoch, the models were captured during the high-velocity phase of structural definition. The regularized Gemma 2 9B run demonstrates this effectively: while its classification ability converged rapidly due to wide multi-head boundaries, its gradient norm remained active (16.39) and its final loss stalled safely at 0.4286, preserving downstream token entropy. Conversely, the distilled DeepSeek geometry maintained a highly conservative reward margin (0.0333), reflecting unique parameter stiffness under DPO transitions.

4 Empirical Dissection of Pathological Overfitted Policy States

When subjected to multi-epoch optimization horizons without strict regularization, each distinct language model architecture manifested specific, repeatable behavioral failure modes during downstream out-of-distribution stress testing. Understanding these failures requires analyzing how non-vanishing gradients systematically break the pre-trained semantic foundations of each model family.

4.1 Overfitted Meta-Llama 3 8B: The Disclaimer Hallucination Mode

The overfitted LoRA adapter for Llama 3 suffered from severe linguistic entropy collapse. Because the model's foundational prior is heavily optimized for conversational helpfulness and safety compliance, the unconstrained DPO gradient pushed the parameters into an over-alignment loop.

When presented with out-of-distribution assets exhibiting high volatility, the overfitted Llama 3 abandoned the concise analytical structure of the training dataset. Instead, it generated excessively verbose outputs dominated by redundant, generic corporate compliance disclaimers (*"Please consult a licensed financial advisor, as investing involves significant risks..."*). This behavioral degradation demonstrates that when a model is overfit via DPO, it fails to synthesize the quantitative matrix and instead falls back on over-generating high-frequency safety text blocks to maximize its token-level reward score.

4.2 Overfitted Google Gemma 2 9B: The Token-Replication Stagnation Mode

Gemma 2 features a unique architecture with a wide attention head dimension (256 dimensions), causing it to converge extremely fast under preference optimization frameworks. However, when exposed to 3 full training epochs, this rapid convergence triggered complete semantic petrification.

The overfitted Gemma 2 policy state locked onto the exact string sequences of the training samples. When stress-tested with completely new stock tickers or distinct numerical parameters (e.g., an asset with a massive -10.5% short-term drop), the model completely ignored the updated inputs. It parroted back the exact, word-for-word textual output associated with the -1.04% training sample. The model's capacity to map data onto logic was entirely lost, replaced by a rigid dictionary-style lookup mechanism that failed to generalize.

4.3 Overfitted Alibaba Qwen 2.5 7B: The Metric Extrapolation Pathology

Qwen 2.5 exhibits high native proficiency at processing data tables and factual technical tokens. Under the influence of preference overfitting, this capability degraded into a severe metric extrapolation pathology.

To maximize the mathematical log-probability gap between the chosen and rejected branches, the overfitted Qwen 2.5 adapter began to invent and interpolate arbitrary scoring scales entirely external to the multi-factor input configurations. It routinely output statements such as *"This asset scores a 2.5 out of 10 on our internal volatility security layout,"* or allocated arbitrary numerical tags like *"Risk Class 4 Evaluation."* This failure mode proves that when overfit, the model attempts to enforce the "preferred" classification boundary by hallucinating field-specific numerical systems to justify its decisions.

4.4 Overfitted DeepSeek-R1-Distill-7B: Cognitive Boundary Shattering

The most severe manifestation occurred within the reasoning-distilled DeepSeek architecture. Because DeepSeek-R1 models are structurally trained via Reinforcement Learning to execute implicit self-correction steps, they rely on a fragile balance between reasoning tokens and final conversational responses.

The end-to-end 3-epoch DPO pipeline shattered these internal cognitive boundaries. Even before completing the second epoch, the overfitted model completely lost control of its output generation boundaries. It bypassed its native XML `<think>` tags, leaking raw internal reasoning streams into the final response text. Furthermore, it suffered from an absolute breakdown of persona tracking, adopting the first-person perspective of the user and entering infinite recursive loops, rehashing the initial prompt queries indefinitely without ever executing the required financial analysis.

5 Mechanistic Analysis: 3-Epoch Collapse vs. 1-Epoch Generalization

The transition from a 3-epoch computational horizon to a disciplined 1-epoch execution path fundamentally alters the optimization dynamics by trapping the model parameters strictly within the **early gradient regime**. Table 2 contrasts the distinct behavioral and operational mechanics of these two optimization states.

Table 2: Operational Trajectory: Early Regime vs. Overfitting Saturation

Phase Variable	Early Regime (Horizon = 1.0)	Saturation Regime (Horizon = 3.0)
Primary Target	Macro-Structural Rule Alignment	Micro-Token Sequence Memorization
Gradient Focus	Conceptual Instruction Following	Log-Probability Maximization
KL Boundary	Stable Policy Boundaries ($D_{KL} < \epsilon$)	Broken Policy Boundaries ($D_{KL} \to \infty$)
Behavioral Tone	Concise, Rule-Anchored, Generalist	Verbose, Hallucinated, Brittle

During a single pass over the preferences, each sample pair is encountered exactly once. The optimization leverages the highest rate of conceptual learning, where the model updates its low-rank matrices (LoRA) to absorb the absolute constraints of the financial prompt (the mandate to ignore euphoria and compute pure risk metrics).

By stopping at epoch 1.0, the training is halted before the optimizer can exploit specific n-gram statistical frequencies to expand the margin artificially. This preserves the structural integrity of the pre-trained weights, effectively bounding the Kullback-Leibler divergence and ensuring that the model retains its open-domain generalization capabilities while incorporating strict risk-averse execution constraints.

6 Systemic Interplay of Regularization Parameters

The hyperparameter recalibration framework utilized a highly synchronized triple-axis brake strategy. The operational mechanics of this parameter interplay are formalized as follows:

1. **Horizon Contraction ($\mathcal{T} = 1.0$):** Acts as a structural constraint, preventing the model from cycling back over solved preference distributions, isolating learning to macro-rules.

2. **Learning Rate Attenuation ($\eta = 1.0 \times 10^{-6}$):** Reduces the parameter step velocity. By scaling down the learning rate, we ensure that the Low-Rank Adaptation matrices cannot execute drastic directional changes in a single batch step, keeping the active policy smoothly bound within the reference model's local loss valleys.

3. **KL Elasticity Expansion ($\beta = 0.15$):** Subitizing β from 0.10 to 0.15 mathematically stiffens the constraint spring. It penalizes token-level deviation from π_{ref}, ensuring that the model retains its structural syntax and formatting capabilities while updating its internal financial sorting logic.

7 Cross-Architectural Behavioral & Inductive Bias Synthesis

A rigorous evaluation across the serialized models using complex, out-of-distribution adversarial matrices exposed distinct cognitive and formatting behaviors dictated by the native inductive biases of each architectural family. Table 3 maps these post-alignment behavioral profiles.

Table 3: Cross-Architectural Behavioral Manifestations Post-Alignment

Model Family	Tone Profile	Analytical Strategy	Downstream Utility
Meta-Llama 3 8B	Empathetic, Conversational	Addresses qualitative queries but enforces data veto	Client-facing interactive advisor
Google Gemma 2 9B	Hyper-Objective, Surgical	Eliminates conversational fillers, outputs direct parameters	High-speed automated execution backend
Alibaba Qwen 2.5 7B	Pragmatic, Structured	Deconstructs data segments, suggests technical tools (stop-loss)	Risk management dashboard integration
DeepSeek-R1 7B	Anomalous, Recursive	Destabilizes identity barriers, leaks internal thinking text	High-logic complex synthesis (requires specialized GRPO structural training)

7.1 Meta-Llama 3 8B Instruct: The Balanced Quantitative Consultant

Post-recalibration, the Llama 3 8B architecture achieved an optimal operational equilibrium, effectively illustrating the preservation of semantic boundaries. The single-epoch training constraint acted as a precision regularizer, updating preference weights without overwriting the heavy alignment anchor originally established by Meta's RLHF safety pipelines.

When exposed to extreme user emotional bias, the calibrated Llama 3 maintained superior conversational prose. It explicitly parsed and addressed the user's focus on social media sentiment, yet immediately decoupled qualitative hype from technical parameters. It executed an absolute, data-driven veto based entirely on the high annualized volatility and the stock's position below the baseline 20-day Simple Moving Average. This structured blend makes it uniquely qualified for wealth management client front-ends.

7.2 Google Gemma 2 9B Instruct: The Surgical and Hyper-Objective Executor

The Gemma 2 9B policy state reached the highest degree of determinism within our experiment. The underlying architecture—featuring wide 256-dimensional attention head projections and explicit logit softcapping layers—ensures rapid parameter updates. Under a constrained 1-epoch horizon, it immediately locked onto the formatting boundaries of the target dataset, completely shedding conversational fillers, preambles, and filler text blocks.

Its post-calibration style achieved near-zero token overhead. When presented with market hype, it slashed emotional narratives from line one, outputting a clinical, bulleted dissection of the statistical matrix and enforcing a strict risk boundary. It operates as a deterministic numerical execution engine, making it perfectly suited for high-speed automated algorithmic trading backends.

7.3 Alibaba Qwen 2.5 7B Instruct: The Structured Risk Engineer

The calibrated Qwen 2.5 7B model excelled at processing complex data matrices and transforming them into defensive asset allocation logic. Rather than outputting a flat, binary refusal string, its low-rank matrices leveraged their strong native inductive bias for parsing variables to systematically isolate individual matrix risks into clean analytical subsections.

Crucially, the 1-epoch regularization prevented it from hallucinating arbitrary scoring metrics. Instead, the model proposed actionable quantitative risk management protocols, suggesting the integration of automated protective stop-loss triggers to insulate the portfolio from the 45.2% annualized volatility. It represents the ideal choice for integration within risk management dashboards and complex corporate BPM execution layers.

7.4 Mechanistic Breakdown of the DeepSeek Identity Anomaly

The behavior displayed by the DeepSeek-R1-Distill-Qwen-7B architecture represents a profound case study in alignment mechanics. Foundational reasoning models undergo extreme Reinforcement Learning focused on generating multi-step computational tokens wrapped within explicit validation boundaries.

When subjected to the end-to-end alignment vectors of our DPO preference matrix, a structural collision occurred. The dataset forced the model to map input directly to output, bypassing extended textual verification. This format broke the model's native cognitive flow.

As captured in the execution logs, the model experienced an identity boundary collapse. It skipped its standard structural XML tags, leaked internal thinking tokens directly into the visible stream, and ultimately projected its operational context onto the user's persona. It entered an unconstrained recursive loop, continually simulating the user's emotional internal monologue rather than executing the assistant's quantitative response. This proves that reasoning-focused configurations are fundamentally brittle when exposed to standard preference optimization scripts, requiring specialized architectural decoupling techniques, such as Group Relative Policy Optimization (GRPO), to maintain structural integrity.

8 Conclusion

This framework demonstrates that the optimization of Large Language Models for automated financial decision-making requires precise control over the mathematical behavior of the implicit reward gradient. By understanding that DPO loss functions continuously push token probability margins even after achieving perfect classification, we successfully implement a triple-axis regularization workflow that bounds the optimization trajectory within an informative 1-epoch regime. The empirical results across the Llama, Gemma, and Qwen geometries demonstrate excellent adaptation stability, successfully removing qualitative and emotional biases from multifactor financial inputs. Future trajectories will explore the construction of unified preference parameters that tie token probability shifts directly to multi-asset covariance matrices, paving the way for adaptive, risk-aware generative trading systems.

9 Declaration of AI Usage

In accordance with the project guidelines, I declare that Artificial Intelligence tools (LLMs) were utilized to format the Python scripts according to clean code standards and to troubleshoot the LaTeX typesetting of this document. All methodological reasoning, experimental design, validation of the predictive architecture, data analysis, and experimental interpretation are original.